1. Six Phases of the Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a process used to plan, develop, test, deploy, and maintain software.

1.1 Planning

The team identifies the problem, project objectives, budget, timeline, resources, and project scope. For RP-Cloud Labs, the team will plan how to provide students and lecturers with browser-based access to learning tools.

1.2 Requirements Analysis

The team gathers and analyses requirements from students, lecturers, college administrators, RP central IT staff, and software vendors. The requirements will include account validation, tool-session requests, and checking college software license capacity.

1.3 System Design

The team designs the system architecture, database, user interfaces, and communication between components. For the Tool-Session Service, the design will show how student requests, license capacity, and session records will be managed.

1.4 Implementation

The team writes the code for the Tool-Session Service using an appropriate programming language and framework. The service will allow students to request tool sessions and check whether their college has available license capacity.

1.5 Testing

The team tests the service to identify and correct errors. Tests will verify session requests, input validation, license-capacity checks, security, and database operations.

1.6 Deployment and Maintenance

The team deploys the tested service using Docker and Kubernetes. After deployment, the team monitors the service, fixes errors, updates features, and improves performance as requirements change.